

Encrypted Protocol Topology Recovery

A Simulation-Only Computational Statistics Laboratory

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Independent Computational Statistics Research

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Author Note. This report covers controlled synthetic evidence only. The primary 1,800-run factorial is complete and provenance-bound; the separate robustness experiment is pending. No external observations enter the software.

Abstract

This report documents a simulation-only computational statistics laboratory for recovering latent directed topology under controlled truth. Version 1.1.0 contains no collection, interface-enumeration, external-trace import, address mapping, or live-inference path. Every event, mark, node label, graph, and truth label used by the runtime is generated in memory from a declared integer seed.

The fitted architecture combines an exact exponential-kernel multivariate Hawkes likelihood, an amortized dynamic stochastic block model, and masked autoregressive variational flows. A static graph autoencoder with Louvain clustering provides the comparator. The likelihood includes both event log-intensities and the exact observation-window compensator; the variational density includes the flow Jacobian; and excitation is constrained below a stability boundary.

The primary $2 \times 3 \times 3 \times 100$ randomized complete-block experiment is complete, producing 1,800 architecture runs. Analyzer v1.0.2 binds inference to the source SHA-256, reports 27 seed-paired contrasts, applies Holm correction within endpoint, and records a diagnostic-aware mixed-model fallback. A separate five-generator, 2,700-run default robustness study is specified but not yet reported. All conclusions are limited to synthetic topology recovery.

Keywords: Hawkes process, stochastic block model, normalizing flow, simulation, topology recovery, model misspecification, negative control

Research Question and Scope

The scientific question is whether continuous-time excitation and flexible latent geometry improve directed graph recovery when synthetic timing and size information are degraded. The estimand is a difference between two algorithms evaluated on the same generated truth graph, not a statement about any real entity or communication system.

Controlled truth separates recovery error from interpretive uncertainty. A simulator supplies a directed binary adjacency matrix and event observations whose stochastic relationship to that matrix is known. Because every design coordinate is explicit, failures can be attributed to architecture, perturbation, sparsity, generator mismatch, or sampling variation rather than an undocumented source.

The design cannot decrypt content, identify a person or organization, infer intent, attribute command and control, or establish operational SIGINT performance. Those quantities are absent from the data-generating process and therefore unidentified by the model. This boundary is enforced in source layout, command-line surface, dependencies, tests, status metadata, and publication artifacts.

Synthetic Observation Space

Events live on the product space of continuous time, directed synthetic dyads, and positive synthetic size marks. The counting measure records occurrences on this space, and the predictable intensity is measurable with respect to the left-continuous event history. Node keys are labels such as `synthetic-node-003` and have no external correspondence.

The event batch contains aligned tensors for times, sizes, sources, and destinations. Validation requires nondecreasing nonnegative times, positive sizes, in-range node indices, and no self-links. These conditions define an auditable mathematical object independent of any external serialization or collection system.

A finite observation window begins at the first generated event. The primary generator and its random-number order remain unchanged from the frozen factorial. New event laws are selected through an explicit registry field and write to a separate result schema, preventing a robustness extension from silently redefining the original experiment.

Exact Multivariate Hawkes Likelihood

For event type r , conditional intensity is a positive baseline plus exponentially decayed contributions from earlier types. The log likelihood on a finite interval equals the sum of observed event log-intensities minus integrated intensity over the full event-type domain. This compensator normalizes the process and cannot be omitted.

With an exponential kernel, the baseline contribution is window length times the sum of baselines. Each prior event contributes its excitation coefficient multiplied by one minus an exponential tail factor. The implementation uses positive smooth transforms, a declared numerical floor, and finite-gradient checks.

Figure 1 shows the sign and role of the three components. The event term rewards intensity at observed events, while both compensator components penalize intensity integrated over exposure. A model that retains only the first term can increase its objective by inflating intensity and does not represent the intended probability law.

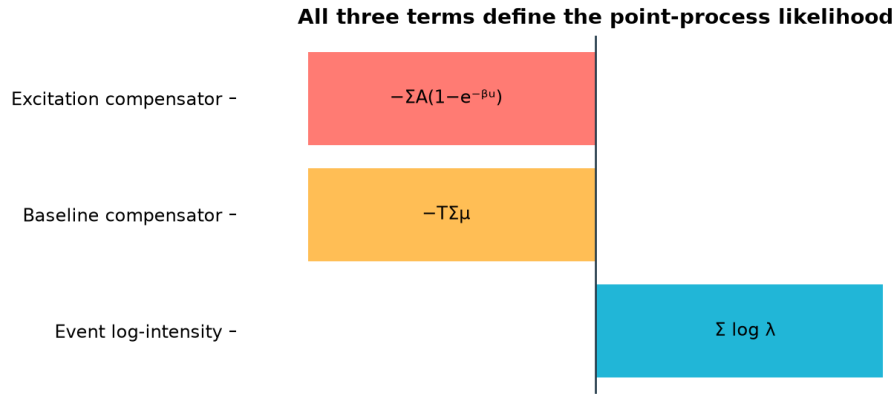


Figure 1. Exact Hawkes likelihood decomposition.

Stability and Partial Identifiability

A stationary linear Hawkes process requires excitation below its critical branching boundary. The software rescales the nonnegative excitation matrix to a spectral-norm ceiling of 0.85. This conservative differentiable constraint keeps the fitted process inside the implemented stability region.

Baseline and excitation are only partially identifiable in short sequences. A high baseline with weak excitation can resemble a lower baseline with stronger history dependence. Block labels also switch without altering likelihood. Evaluation therefore uses permutation-invariant link probabilities and never attaches semantic meaning to a block number.

The robustness registry makes these limitations testable. A mixture kernel probes time-scale mismatch, a Cox law introduces shared rate changes, a renewal law creates non-Poisson intervals without self-excitation, and a null law removes topology information. Agreement under one generator is not treated as universal identification.

Synthetic Marks and Perturbations

Synthetic size is modeled as a conditional mark associated with the generated block relation. The primary event-time likelihood remains a multivariate counting process. This factorization declares that, conditional on latent structure and dyadic type, the size contribution does not alter the event-time compensator.

The padding condition maps all observed synthetic sizes to 1,500 units and injects declared dummy events. The jitter condition adds seeded Gaussian displacement to generated times, clips to the window, and stably reorders observations. The none condition retains the generated values. Truth adjacency remains unchanged across perturbations.

These perturbations are abstractions rather than copies of a field system. They test loss or contamination of model inputs within a controlled experiment. Generalization is limited to the coded mechanisms and parameter values, so any stronger robustness statement requires additional registered laws and a new frozen design.

Dynamic Stochastic Block Structure

The block model represents each synthetic node by a probability vector over a finite truncation of latent roles. Directed block-pair logits generate asymmetric link probabilities. Relaxed Gumbel-Softmax draws make discrete allocations differentiable during optimization while allowing unused blocks to receive negligible mass.

Node features summarize generated outgoing and incoming counts, size moments, and timing rates. An amortized encoder maps these features to variational parameters. The resulting embeddings enter both relaxed block allocation and the directed link decoder.

Blocks are structural abstractions, not identities or categories. Their labels can exchange under posterior symmetry. The defensible object is a posterior over synthetic relational structure and occupied components, evaluated against the simulator's binary adjacency matrix.

Masked Autoregressive Variational Flow

A diagonal Gaussian posterior may be too rigid for graph-allocation uncertainty. A masked autoregressive flow transforms Gaussian base draws through triangular bijections. Each coordinate shift and scale depends only on preceding coordinates under the selected ordering, yielding a tractable Jacobian.

Bounded log scales prevent numerical explosion while preserving invertibility. Alternating coordinate order improves dependence coverage. Forward and inverse tests require reconstruction within numerical tolerance and cancellation of the corresponding log determinants.

The flow increases approximation flexibility but does not make the posterior exact. Importance effective sample fraction describes degeneracy of variational importance weights only. It is not Markov-chain effective sample size and is not used as a cross-architecture endpoint.

Evidence Lower Bound and Optimization

The evidence lower bound is the Monte Carlo expectation of joint log density minus variational log density. The joint term combines the exact event-time likelihood, synthetic mark contribution, block-structured graph prior, and regularization. The variational term includes Gaussian base density and every flow log Jacobian.

Optimization uses Adam, gradient clipping, seeded initialization, and finite-value guards. Device synchronization surrounds measured compute segments when acceleration is available. The final objective per event is a training diagnostic rather than a confirmatory architecture score because the two methods optimize different objectives.

Variational restarts and epochs are not Markov chains. The project therefore does not compute Gelman-Rubin statistics from optimization histories. Rank-normalized folded R-hat remains appropriate only for future algorithms that produce genuine independent chains.

Static Comparator

The comparator aggregates directed synthetic event counts into a static graph, encodes node features with a graph autoencoder, and decodes link probabilities from the latent representation. Louvain partitioning supplies an occupied-community count for descriptive comparison.

It deliberately lacks continuous-time intensity and cannot represent event-history excitation. That limitation defines the architecture contrast: whether the additional temporal likelihood and flexible posterior improve link recovery enough to justify their computational cost.

Both methods receive the same generated batch and truth labels within a seed and condition. Identical scoring code computes off-diagonal link AUC and binary link log score. Latency is normalized by event count; architecture-specific diagnostics remain separate.

Simulation-Only Software Boundary

The executable interface contains only simulation-registry, simulate, run-factorial, analyze, run-robustness, and analyze-robustness. There is no interface enumeration, acquisition backend, external event-file argument, address transformer, or live-inference command.

The package exposes EventBatch but no persistent external-event record. Packet-oriented dependencies are absent. Boundary tests assert that the former acquisition, storage, and pseudonymization modules cannot be imported and that the command set contains exactly the six approved operations.

This boundary is a scientific design choice. It eliminates ambiguous provenance and confines all link-truth statements to known simulation. Suitable extensions include new stochastic generators, priors, perturbations, calibration procedures, posterior-predictive checks, and compute experiments.

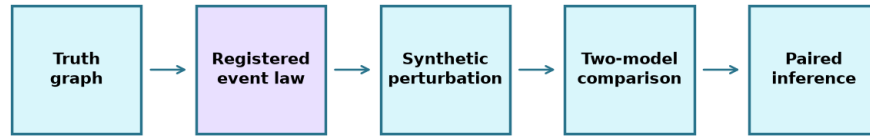
Registered Evidence Path

Figure 2 summarizes the only runtime evidence path. A seed produces a truth graph and block memberships. One registered law produces an event batch. A declared perturbation changes observations without changing truth. Both architectures receive that batch, and inference operates on paired seed differences.

Registry metadata records name, temporal process, presence or absence of topology signal, fitted-model alignment, and scientific purpose. This metadata is emitted through the CLI and embedded in robustness summaries so a numerical result remains attached to its data-generating assumptions.

All five generators return the same validated object shape. This common contract enables matched scoring while preventing informal substitution of an undocumented data source. Unknown generator names fail before random generation begins.

Simulation-only evidence path



All events, node labels, marks, graphs, and truth labels are generated in memory

Figure 2. Simulation-only evidence path.

Matched Exponential Hawkes Generator

The primary generator draws balanced node memberships and a directed block-connectivity matrix conditional on at least one absent off-diagonal block edge. This condition guarantees both classes for link AUC without editing a degenerate draw after the fact.

Block connectivity controls immigrant rates, while diagonal and row-or-column-related excitation terms create event dependence. A branching construction draws immigrant events and recursive offspring under an exponential delay law. Synthetic node pairs and sizes are then sampled within the event's source and destination blocks.

Version 1.1.0 retains this generator as the default and preserves its random-number sequence and parameters. Explicit and implicit selection of `hawkes_exponential` produce identical tensors. This compatibility rule protects the frozen primary factorial from retrospective change.

Misspecification Generators

The mixture-Hawkes law uses short- and long-scale offspring delays. It preserves excitation and topology signal but lies outside the fitted single exponential kernel. Performance loss isolates sensitivity to kernel-shape mismatch.

The piecewise-Cox law applies shared lognormal multipliers across six time segments. It preserves topology-linked baseline differences while adding external rate shocks that may resemble event dependence. The gamma-renewal law generates independent mark-specific renewal sequences with shape below one, creating bursty intervals without self-excitation.

These families do not span every possible event process. They form named, reproducible probes of three failure modes: kernel scale, nonstationarity, and dependence mechanism. Conclusions must name the generator rather than use an unrestricted claim of robustness.

Independent No-Signal Negative Control

The independent-null family draws homogeneous Poisson events with equalized mark rates unrelated to block connectivity. A binary truth graph is still drawn so the same evaluation functions run, but observations contain no recoverable topology signal by construction.

Its purpose is calibration. An architecture that systematically reports strong discrimination under this law may exploit finite-sample artifacts, scoring leakage, optimization bias, or an unintended relationship in the simulator. Near-chance discrimination is the desired behavior.

The null does not prove absence of a real relationship and cannot validate a decision about any entity. It is solely an internal negative control for the synthetic pipeline. Its registry record explicitly declares `truth_signal` false.

Frozen Primary Factorial

The primary design crosses two architectures, three perturbations, and three sparsity levels within each of 100 seeds, for 18 cells per seed and 1,800 runs. Each seed is a complete block because it supplies every treatment combination.

Within-block order is deterministically shuffled. Shared seeds produce paired architecture contrasts and reduce between-graph variation. Primary endpoints are link AUC, binary link log score, and compute latency per event. Failures remain part of the design status rather than being silently discarded.

Figure 3 shows the nine perturbation-by-sparsity conditions and the two distinct experiment sizes. The robustness extension uses its own generator-prefixed schema; it cannot append to the frozen primary table.

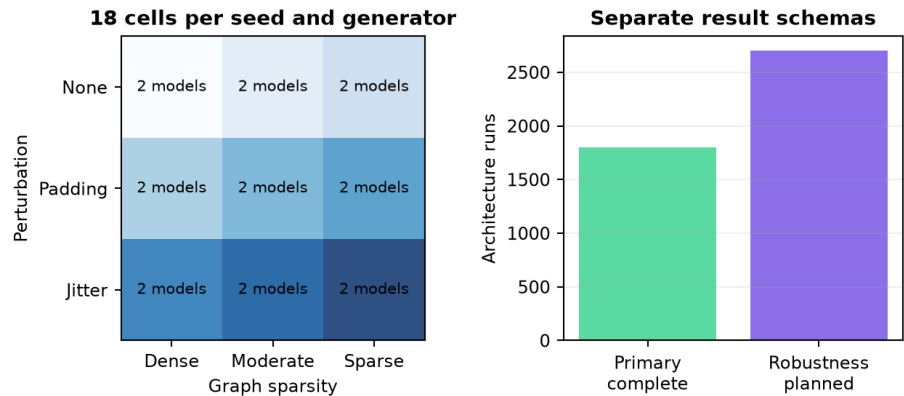


Figure 3. Primary and robustness allocations.

Completed Primary Evidence

The primary design is complete. Its source artifact is identified by SHA-256 d19433ff32f2222655f5408810b9ed6fd59fa01ea461e05e3f1aa5a59977309a. Version 1.1.0 does not modify or rerun any of its 1,800 cells.

Across nine conditions, the Hawkes-flow DSBM minus static link-AUC contrast ranges from -0.1542 to -0.0792, with all nine Holm decisions rejecting at 0.05. Link log-score contrasts range from -0.0200 to +0.0064, with seven of nine Holm decisions rejecting. Latency contrasts range from +11.4148 to +15.2479 milliseconds per event, with all nine rejecting.

These ranges show that greater model complexity did not yield superior AUC in the matched primary study and incurred higher latency. Sign and scale must be interpreted through the documented endpoint definitions. The result is evidence about this synthetic generator and configuration only, not real systems.

Analyzer v1.0.2 Inference

The analyzer computes the result CSV SHA-256 before and after reading and aborts if the file changes. It requires 1,800 unique completed keys before publishing a summary. Every condition-specific architecture contrast is calculated from seed-level paired differences.

For each endpoint, nine paired two-sided tests receive Holm step-down adjustment controlling familywise error at 0.05. Confidence intervals are condition-specific unadjusted 95% intervals. The summary records raw and adjusted p values, decision flags, standard errors, and paired seed counts.

The factorial model first attempts a seed random intercept plus architecture slope, then a random intercept only. Inadmissible mixed fits fall back to the same fixed design with seed-clustered OLS inference. Convergence warnings, covariance validity, and every rejection reason are retained rather than hidden.

Robustness Experiment

The robustness design crosses five generators with two architectures, three perturbations, three sparsity levels, and 30 default seeds. This yields 2,700 cells. Output is restartable and keyed by seed, generator, architecture, perturbation, and sparsity.

Analysis requires the exact expected key set and complete status. Within every generator and condition, architecture differences are paired by seed. Each endpoint has 45 generator-by-condition comparisons, which receive one Holm correction family at alpha 0.05.

The analyzer produces 135 contrasts across three endpoints, preserves the verified input hash, and embeds selected registry records in its JSON summary. Because the experiment is pending, this report gives no cross-generator numerical performance claim.

Convergence and Calibration Diagnostics

Optimization checks include finite objectives, gradient clipping, occupied components, posterior entropy, and importance-weight effective sample fraction. These diagnose numerical behavior but do not prove posterior correctness.

Generator-specific posterior-predictive checks should compare event counts, interarrival distributions, size quantiles, and synthetic dyad counts. Simulation-based calibration can be added when the full generative prior and posterior targets are aligned. The independent null directly audits spurious link discrimination.

Static and dynamic failures are counted by condition. Complete-case filtering is prohibited because selective success could bias model comparison. Figure 4 distinguishes completed mathematical and primary evidence from pending robustness and independent replication.

Compute and Latency

The confirmatory latency endpoint measures fitting compute and divides by generated event count. Device synchronization is used where applicable. Event generation and result-file operations are excluded because the estimand is architecture compute, not end-to-end system throughput.

Hardware, software versions, device type, thread configuration, and warm-up policy should accompany every latency result. Cross-device comparisons require separate strata or calibration. Acceleration is optional and a CPU result remains valid for its declared environment.

The completed primary contrasts show a substantial positive Hawkes-flow-minus-static latency difference. The robustness design asks whether this cost changes with event law and perturbation. It does not make a real-time, operational, or streaming-performance claim.

Software Verification and Provenance

The permanent suite spans Hawkes likelihood and stability, flow inversion, model fitting, simulator determinism, factorial inference, generator registry behavior, exact robustness locks, and the simulation-only boundary. Continuous integration installs a CPU PyTorch wheel, lints, tests, rebuilds this report, and verifies publication claims.

Experiment output is restartable and written atomically. Primary and robustness schemas are distinct. Public summaries bind aggregate inference to exact source hashes. The machine-readable status distinguishes the completed 1,800-cell factorial from the pending 2,700-cell robustness suite.

Generated PDF and dashboard artifacts remain in the repository for inspection, while source scripts are authoritative. An exact 27-page check prevents accidental report drift. Passing software verification demonstrates implementation consistency, not new empirical evidence.

Evidence gates preserve completed and pending status

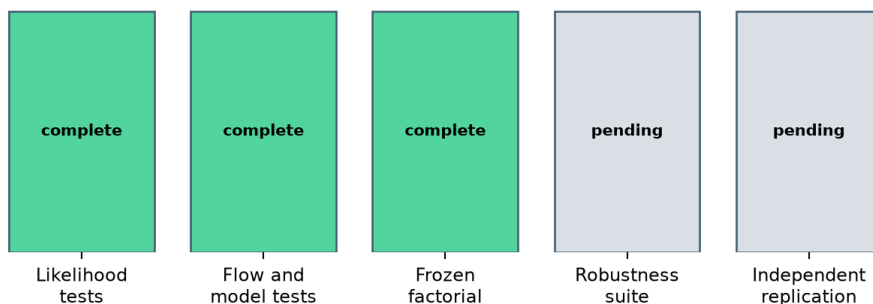


Figure 4. Completed and pending evidence gates.

Threats to Validity

Internal validity depends on correct truth generation, deterministic allocation, stable dependencies, and complete failure reporting. The primary generator is close to the fitted model, which can favor the dynamic architecture in principle even though the observed AUC contrasts favored the comparator.

Construct validity is limited to binary synthetic block connectivity and the implemented event laws. Link AUC and log score quantify recovery of that adjacency, not semantic relationships. Padding and jitter are narrow perturbations and should not be treated as exhaustive degradation mechanisms.

Statistical conclusion validity depends on seed pairing, multiplicity control, admissible covariance inference, and no outcome-driven changes. External validity is intentionally restricted to the registry. Independent replication and additional preregistered generators are needed before broad algorithmic claims.

Discussion and Conclusion

The project now provides a falsifiable simulation laboratory with an exact point-process target, invertible variational density, controlled graph truth, deterministic generator registry, static comparator, provenance-locked inference, and explicit negative control.

The completed primary study shows that additional temporal and variational complexity did not improve link AUC under the matched design and carried a clear compute cost. That result is scientifically useful: sophistication is not assumed to imply superiority. The pending robustness suite asks whether the architecture ranking changes under declared misspecification.

The strongest defensible conclusion remains narrow. This software evaluates synthetic topology recovery under controlled stochastic assumptions. It does not ingest real observations and supports no conclusion about encrypted content, identity, intent, attribution, command and control, or operational SIGINT performance.

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References define the point-process, block-model, flow, graph-comparison, and multiplicity methods implemented in the repository.

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Artifact	Verification
Runtime boundary	Six simulation and analysis commands only
Primary factorial	1,800 complete cells; analyzer v1.0.2
Source provenance	d19433ff32f2222655f5408810b9ed6f d59fa01ea461e05e3f1aa5a59977309a
Robustness suite	Five generators; 2,700 default cells pending
Report	Generated from source; exactly 27 pages